LESSON PLAN · SCRATCH (GR 3–5) BEGINNER: SCRATCH GAME MAKERS

Lesson 2: Events: Make Things Happen When...

Module 1 | 45 minutes | Events | Grades 3–5 · ages 8–10

Objective: Use event hat blocks (green flag, key pressed, sprite clicked) to start scripts, and control a sprite with the arrow keys.

Success criteria

- I can use a hat block to start a script.
- I can make a sprite move with the arrow keys.
- I can explain why a script without a hat doesn't run.

Key vocabulary

- **event**: Something that happens, like a key press or a click.
- **hat block**: A block with a round top that starts a script when its event happens.
- **x position**: How far left or right a sprite is.
- **y position**: How far up or down a sprite is.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Event game: clap/stamp/"Scratch!". Link to hat blocks.
5 min	**Watch**	Lesson video. Pause on "No hat, no start" and ask why the left script never runs.
13 min	**Build along**	Arrow-key controls, start position and click reaction in Scratch.
10 min	**Perform**	Start me up puzzle, matching, pick-the-script and sorting activities.
7 min	**Probe**	Space-to-go puzzle, debug Ria's script, and the game-events writing task.
5 min	**Wrap-up**	Exit ticket; volunteers drive their sprite on the projector.

Unplugged warm-up

Event game (5 min): agree three events — "clap" = stand up, "stamp" = sit down, "say Scratch!" = wave. Call events randomly. Point out that students waited for their event, just like scripts.

Teaching the idea

Event-driven programming: scripts wait for an event (flag, key, click, message) and then run. A hat block names the event. Without a hat, a script never starts by itself. One sprite can have many scripts with different hats. x is horizontal position, y is vertical.

Analogy: A hat block is like a doorbell: nothing happens until someone presses it — then the right person answers.

Live demo: Build the right-arrow script live. Then remove the hat and press the arrow: nothing happens. Put the hat back. Then add the left arrow but "forget" the minus sign and let students debug it.

Questions to ask

Ask	Listen for
What event starts this script?	The one written on its hat block.
Why didn't anything happen?	The script had no hat block, or a different event was used.
How do we move left?	Change x by a negative number.

Differentiation

Support
Give a printed card showing the four arrow-key scripts. Students build only left and right first.

Stretch
Add "when [w] key pressed" etc. so a second player can use W/A/S/D, or make the sprite say where it is using the x position reporter.

Common misconceptions

Misconception	How to fix it
A script without a hat runs when I press the flag	Only scripts with "when green flag clicked" on top run when the flag is clicked.
change x by (10) moves the sprite 10 squares	It moves 10 small steps (pixels) — the stage is 480 steps wide.
Up is change x	x is left–right; y is up–down.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

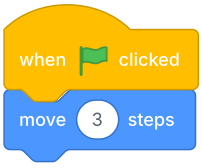
Exit ticket
Finish the sentence: “A script with no hat block…”. Then write the block that moves a sprite left.

Homework
Find three events in the world around you (a doorbell, a traffic light…). Write what happens WHEN each event happens.

Answer key — Kitabash activities

1. Start me up (perform · block code)

One solution:

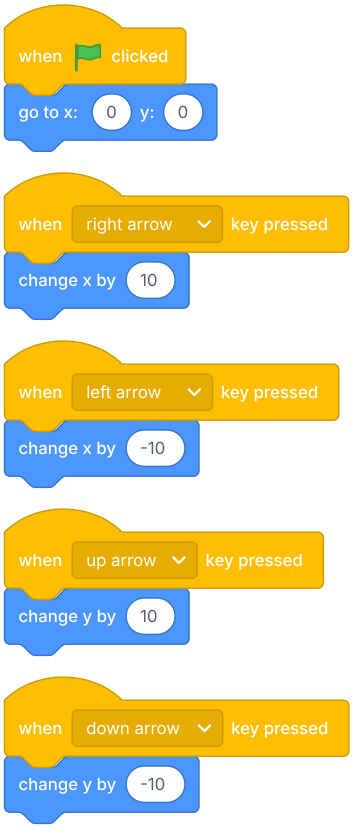


2. Code Lab: drive with the arrow keys (perform · code lab)

Goals checked: → moves right · ← moves left · ↑ moves up · ↓ moves down · Flag puts Pip in the middle

One solution:

Pip

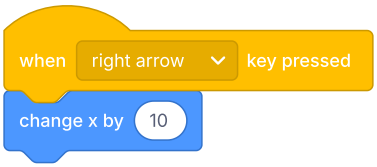


3. Match each hat block to when its script runs (perform · matching)

when green flag clicked → When the green flag above the stage is clicked · when space key pressed → When you press the space bar · when this sprite clicked → When you click on that sprite · when backdrop switches to night → When the stage picture changes to “night”

4. Pick the script (perform · multiple choice)

Answer: A

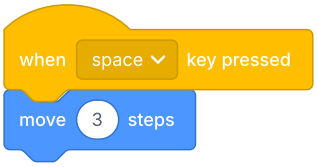


5. Hat block or action block? (perform · categorize)

Hat block (starts a script): when green flag clicked, when this sprite clicked, when space key pressed
Action block (does something): move (10) steps, say Hi! for 2 seconds, change x by (10)

6. Space to go (perform · block code)

One solution:



7. Debug it (probe · multiple choice)

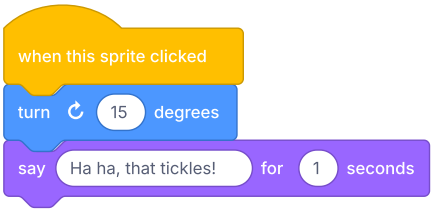
Answer: Change 10 to -10

8. Code Lab: click to react (probe · code lab)

Goals checked: Uses “when this sprite clicked” · Pip reacts to a click · Pip says it tickles

One solution:

Pip



9. Explain your thinking (probe · free response)

Look for: when, press, click. **Example:** In my racing game, when I press the up arrow, the car speeds up. When I click the pause button, the game stops.

Answer key — worksheet

1. **What starts it?:** Script 1: when the space key is pressed. Script 2: when the sprite is clicked.

2. **Left or right?:** RIGHT · LEFT · UP · DOWN
3. **Spot the bug:** There is no hat block. Add “when green flag clicked” on top.
4. **Design a control:** “when space key pressed” and “change y by (50)” (any positive number).

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.